

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Formative Assessment 3

Formative Assessment (FA-3) - Answer Key

Class: Class 8

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL - Database Fundamentals

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 x 1 = 5 marks)

Q.No	Answer	Explanation
1	b	SELECT is used to retrieve data from a database
2	c	WHERE clause filters rows based on conditions
3	c	DELETE FROM students WHERE roll_no = 5 deletes only the row where roll_no = 5
4	c	VARCHAR is used for variable-length text strings like names
5	b	DROP TABLE removes the table structure and all data permanently

Section II: Short Answer Questions (5 x 2 = 10 marks)

1. Write the SQL command to create a table Books.

• **Answer:**

```
CREATE TABLE Books (  
    book_id INT PRIMARY KEY,  
    title VARCHAR(100),  
    author VARCHAR(50),  
    price DECIMAL(10, 2),  
    stock INT  
);
```

• **Marking:** 1 mark for correct CREATE TABLE syntax, 1 mark for all columns with data types.

2. Differentiate between DELETE, TRUNCATE, and DROP commands.

• **Answer:** DELETE removes specific rows with WHERE clause and can be rolled back. TRUNCATE removes all rows but keeps structure, faster than DELETE. DROP removes the entire table including structure.

• **Marking:** 1 mark for each correct differentiation point (3 points for 2 marks).

3. Write SQL queries for insert, update, and delete operations.

• **Answer:**

```
-- Insert 3 records  
INSERT INTO Books VALUES (1, 'Python Basics', 'John Smith', 450.00, 25);  
INSERT INTO Books VALUES (2, 'Web Development', 'Jane Doe', 550.00, 15);  
INSERT INTO Books VALUES (3, 'Database Systems', 'Bob Wilson', 600.00, 10);  
  
-- Update price where book_id = 2  
UPDATE Books SET price = 499.00 WHERE book_id = 2;  
  
-- Delete where stock = 0  
DELETE FROM Books WHERE stock = 0;
```

• **Marking:** 1 mark for correct INSERT statements, 1 mark for correct UPDATE and DELETE.

4. **What is a Primary Key? What are its constraints?**

- **Answer:** A Primary Key uniquely identifies each record in a table. Constraints: Must be unique (no duplicate values), cannot be NULL, each table can have only one primary key.
- **Marking:** 1 mark for definition, 1 mark for constraints.

5. **Write a SQL query to display books sorted by price descending, priced above ₹500.**

- **Answer:**

```
SELECT * FROM Books
WHERE price > 500
ORDER BY price DESC;
```

- **Marking:** 1 mark for WHERE clause, 1 mark for ORDER BY clause.

Section III: Long Answer Questions (5 x 1 = 5 marks)

1. **Write SQL queries for the Students table.**

- **Answer:**

```
-- Display students who scored more than 80 marks
SELECT * FROM Students WHERE marks > 80;

-- Count students in each section
SELECT section, COUNT(*) as student_count
FROM Students
GROUP BY section;

-- Find max, min, and average marks
SELECT MAX(marks) as maximum, MIN(marks) as minimum, AVG(marks) as average
FROM Students;

-- Display students sorted by marks descending
SELECT * FROM Students ORDER BY marks DESC;

-- Show only top 5 students
SELECT * FROM Students ORDER BY marks DESC LIMIT 5;
```

- **Marking:** 1 mark for each correct query (5 queries).

Part B: Hands-on Practical Lab Task (5 marks)

SQLite Database Program:

- **Answer:**

```
import sqlite3

# Create in-memory database
conn = sqlite3.connect(':memory:')
cursor = conn.cursor()

# Create Employees table
cursor.execute('''
    CREATE TABLE Employees (
        emp_id INTEGER PRIMARY KEY,
        name TEXT NOT NULL,
        department TEXT,
        salary REAL,
        joining_date DATE
    )
''')
```

```

# Insert 8 records
employees = [
    (1, 'Alice', 'IT', 75000, '2021-03-15'),
    (2, 'Bob', 'HR', 65000, '2020-07-20'),
    (3, 'Charlie', 'IT', 80000, '2019-01-10'),
    (4, 'Diana', 'Finance', 70000, '2022-05-25'),
    (5, 'Eve', 'HR', 68000, '2021-11-30'),
    (6, 'Frank', 'IT', 72000, '2020-09-05'),
    (7, 'Grace', 'Finance', 85000, '2018-12-01'),
    (8, 'Henry', 'Marketing', 60000, '2023-02-14')
]
cursor.executemany('INSERT INTO Employees VALUES (?, ?, ?, ?, ?)', employees)

# Query 1: SELECT with WHERE
print("IT Department Employees:")
cursor.execute("SELECT * FROM Employees WHERE department = 'IT'")
for row in cursor.fetchall():
    print(row)

# Query 2: ORDER BY
print("\nEmployees sorted by salary:")
cursor.execute("SELECT name, salary FROM Employees ORDER BY salary DESC")
for row in cursor.fetchall():
    print(row)

# Query 3: GROUP BY
print("\nEmployees per department:")
cursor.execute("SELECT department, COUNT(*) FROM Employees GROUP BY department")
for row in cursor.fetchall():
    print(row)

# Query 4: Aggregate functions
print("\nSalary statistics:")
cursor.execute("SELECT AVG(salary), MAX(salary), MIN(salary) FROM Employees")
for row in cursor.fetchall():
    print(f"Average: {row[0]:.2f}, Max: {row[1]}, Min: {row[2]}")

# Query 5: UPDATE
cursor.execute("UPDATE Employees SET salary = salary * 1.1 WHERE department = 'IT'")
conn.commit()

conn.close()

```

- **Marking:** 1 mark for database creation, 1 mark for table creation, 1 mark for inserting records, 1 mark for queries, 1 mark for output formatting.

Part C: Notebook Maintenance (5 marks)

- **Neatness and organisation of notebook (2 marks):** Check for clean, organized notebook.
- **Completion of all assigned classwork and homework (2 marks):** Verify all work is complete.
- **Proper headings, dates, and indexing (1 mark):** Ensure proper formatting throughout.